

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,®ç®±i¼šhs3434@foxmail.com -
ä, ºäºç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖â❖ “**i¼šAMD ROCm Python â❖Žç«`â¼â❖ ‘â • ¥ç`â, ^i¼^ROCm Radeon Cloud
æ-¹â❖ ‘i¼% --- ## ä, ºäºç®€ä» 3 â¹´ç” Ýç%©äzjæ❖`â • ¥ç`âÆ-ç»❖éâEi¼Æè ± ââè½-âž`
K8s â❖Žç«`æ ^â€, \*\*â®¹â™™âÆ-éf`ç½²â€❖èµ,,æº❖èºfâº|â€❖ä»»âŠiç¼-æž’ â€❖GPU
æž`ç❖ + æœ❖âŠi\*\*â€, æœ€èz ‘ 1 â “è ± ââ| K8s / Helm i¼Æç«-ç«`â®Ææ^❖ [**k8s-llm-
runtime**](https://github.com/hs3434/k8s-llm-runtime)i¼^K8s ä, Šçš,, vLLM
æ`jâž`æœ❖âŠiç½ ‘â...³i¼%i¼Æ**v0.1.0 tagged i¼Æâ❖`çž`âœºè • ‘demo**â€, ç + Ýæ,%
Snakemake ç±» Workflows çš,, ML â¹³â❖ºæŠ½è±jâ€, ## æš€æœ-â❖šâ®ç - [Transformer
â»žâ...¥è—`â^ºä, Šæ%«](https://hs3434.github.io/2025/03/03/transformer1/) - [GPT éžžæ ¼
Transformer è Š £ç ❖ EEGi¼šä, ºä»æ¹^â€❖æ€žâ¹^â❖šâ€❖è, ©âº + â»æ¹^â❖ ‘](https://
hs3434.github.io/2025/06/08/transformer-eeg-decoding/) - [â»žžé¶ æ❖â, €â¥—ç«`â^ºç«` EEG
âzjâ❖ • â¶,ç❖ + â • ¥â... • é “¾](https://hs3434.github.io/2025/06/08/eeg-signal-processing-
toolchain/) --- ## æ±,è❖Ææš€èf½i¼^æÆ%â²—â½❖âÆ¹é...❖æž’ âº❖i¼% | ç±»â^« |
æš€èf½ | |-----|-----| | **â❖Žç«`æ , âzj** | Python 3.11+â€❖FastAPIâ€❖pytest | |
**â®¹â™™â, Žç¼-æž’ ** | Dockerâ€❖Kubernetesâ€❖Helmâ€❖é • œâf❖â» “âº “çºiç❖ + | |
GPU / LLM ç” Ýæ€❖ | vLLMâ€❖PyTorchâ€❖Hugging Face | | **â • ¥â... • é “¾** |
uvâ€❖mypyâ€❖GitHub Actions | | **èzç»`\*\* | Linux èzç»` | | **æ • ºæ❖âº “** |
PostgreSQLâ€❖MySQLâ€❖Redis | --- ## èj¹çºç»»éâE ### k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM
æ`jâž`æœ❖âŠiç½ ‘â...³ - è ± ââ|demoéj¹çº **è—®éç`~**i¼^â¹âº” JD è❖Æè´£ 1 +
3i¼%i¼švLLM æ`jâž`éf`ç½²éœ GPU + â®¹â™™ + è • `ç” ±i¼Æâºçé~Ýâ¶šâºâ...±âºæ—
¶ æ`jâž`çºiçº + âºæ❖, i¼ç” ç©¶ /âºâºçé~Ýæf³ç” ” OpenAI â...¼â®¹ API æ— ¶ é—
” æ Š é«~â€, i¼^çº æžžâ¹âº” JD i¼š”â¶ Š æ`jâž`èç»»f/æž`ç❖ +
Workshopâ€❖æ • ™ç`â€❖âšžèf½æ¼” çº”i¼% **æž ¶ æž,,**i¼^â❖Æ chart + ä, %â±, lib +
é • œâf❖â¼^âÆ-i¼%i¼š - `charts/llm-inference/`i¼švLLM è`Ýè½½ charti¼Æ**GPU vendor
â^ ± æ❖çâ»...é❖ `values.yaml` ** i¼^amd.com/gpu / nvidia.com/gpu / none
ä, %æ€i¼Ævalues æ~¾â¼â❖é€%æ©i¼% - `charts/llm-router/`i¼šFastAPI ç½ ‘â...³ chart -
RBAC æœ€âº❖æ❖fé™❖i¼^Pod/Service/ConfigMap read+write + Leasei¼% - `src/
k8s_llm_runtime/`i¼š3 â±, lib æš½è±j - `K8sJobOperator`i¼^â½žâ±, i¼Æéšç” ” Job
èºfâº|i¼% - `VLLMInferenceOperator`i¼^â, â±, i¼Æhelm CLI subprocess + GPU vendori¼% -
`ModelOperator`i¼^é«~â±, i¼ÆOpenAI â...¼â®¹ + è ± âšš`éf`ç½²i¼% **4
â¶ Š æš€æœ-âºç, **i¼š 1. **lib ä, %â±, â^ + â±, æš½è±j**i¼^â¹âº” JD
â❖Žç«`æž ¶ æž,,èf½âšž i¼% - ä, Šâ±, ModelOperator â❖ââ...³âzj load_model(alias) /
unload_model(alias) - ä, â±, VLLMInferenceOperator âºè£... helm ç» + èš, - â½žâ±
, K8sJobOperator éšç” ” K8s Job èºfâº| 2. **Helm ConfigMap æ³”â...¥ aliases**i¼^â¹âº” JD
â®¹â™™âÆ- + èµ,,æº❖èºfâº|i¼% - é❖ââ...❖æ-❖æ-ºâçžæ`jâž`éf½ rebuild
é • œâf❖i¼^GPU é • œâf❖ 15-20GB rebuild 5-10mini¼% - ConfigMap
ç Š ´ çº Š ç” Ýæ • ^i¼Æé¶ ¶ é • œâf❖é ± ❖âº 3. **K8s Lease â + â, fâ¼❖é” ❖**i¼^â¹âº”
JD é + ç¾èzç»` + HAi¼% - coordination.k8s.io/v1 Lease é²æç Router
â¶šâº-æœ-â¹ ¶ â❖ ‘éf`ç½²â❖Æä, €æ`jâž`- é” ❖ç²’ âºæÆ% model
aliasi¼Æè ± âšš`èz ± æœÝ + ç»çº| 4. **ä, šâŠiçº Š Prometheus metrics**i¼^â¹âº” JD
â¹³â❖â❖-è Š, æµæ€ Š i¼% - `chat\_completions\_duration\_seconds`i¼^histogrami¼Æâ❖«
status code æ ± ç¾i¼% - `http\_requests\_total`i¼^counteri¼ÆæÆ% method/route/statusi¼%
- `models\_loaded`i¼^gaugei¼Æâ❖æ~ in-memory statei¼% **GPU vendor â^ ± æ❖ç
demo**i¼^â¹âº” JD âšž â^ + GPU é + ç¾èi¼%i¼š - values.yaml `gpu.vendor=amd|nvidia|
none` - AMDi¼šamd.com/gpu èµ,,æº❖èºfâº|i¼ÆROCm vLLM é • œâf❖ - NVIDIAi¼šnvidia.com/
gpu èµ,,æº❖èºfâº|i¼ÆCUDA vLLM é • œâf❖ - nonei¼šCPU only æ`jâ¼❖i¼^kind cluster

é»è®i¼Ææœ¬æ°ä¼äâ ‘ç’ i¼% **è´´é± ä¿è- **i¼š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint â... clean - 2 Helm chartsi¼^llm-inference + llm-routeri¼% **è, ©â ‘ä, žä¿â®**i¼š - kind èš,ç,¹â½â + ...

é • œâfâš éëYi¼šcontainerdConfigPatches + DaoCloud mirror ### âY°ä°ž Snakemake çš,, ML â¹³â°â • ¥ä½œæµæ; + æž ¶i¼^ç” Yä¿¿èfÆæ™i¼% - èfÆæ™i¼šSnakemake rule è - æ³ • æ— æ³ • âžçžæ”jâ—ésæ,,ç»,â^ - è¾è®i¼šæ “â¼f rule è - æ³ • i¼Æç’ æž¥è°fâ + ...éf” æž¥â°£âš” ææç” Yæ^ rulei¼Æèš,,èÆfè¾ “â... ¥è¾ “â±°ç±»âžâ°³â°çš - æœ” â¼æç¼æž¥ - â...³è” i¼šJD âš â + éj¹æœ°â™”â - |ä¹â¹³â°ç” â ‘ç»é°Æ”â€” â€” æœ¬éj¹ç»æ” ML â¹³â°æš½è±jçš,,â • ¥ç”â°žè • µ ### è,, ‘æœ°æž¥â°£ CNN èš £ç â™”i¼^PyTorch i¼%â€” â€” è±â¿çš,,demoéj¹ç® - ç«â°ç« - æ— ¶ç©â°â°çš - çš - èš £ç EEG ä¿jâ° - - PyQt â°žæ— ¶ GUI + FastAPI æš´éœ² - **â» “â° ***i¼šgithub.com/hs3434/bci-pipeline-demo --- ## â • ¥ä½œç»âž + ### ä, šæµ • â°

æè±Æâ¿jæ°çš ‘æšæœ%é™â...¬â°, i½œç” Yç%©ä¿jæ°â • ¥ç”â, ^i½œ 2025.04 - è±³äš - **rnaseq**i¼^éj¹ç»ä, »âš i¼%oi¼šâY°ä°ž Snakemake çš,, RNA-seq è±âš”âÆ¬â + æž pipelinei¼ÆDocker / Apptainer â¹â™”âÆ¬i¼Æ20 æ, â¹ ¶è¿i¼>config.yaml é©±âš”é...ç½°i¼>â...¬â°, ä, šâš¿è±âš”âÆ¬ä, »ç°¿ - **helix**i¼^Python â°žç« - APIi¼%oi¼šFastAPI + Click + Pydantic v2 â°žçž°çš,,ç” Yç%©ä¿jæ° API æœâš¿i¼Æç»™rnaseq pipeline æ°°¾ç» “æžœâ± • ç°â±, i¼>ç±»âžâÆ¬ + æµ«è - èj + ç» - **æœ°âš¿ç«è¿ç»´**i¼šLinux æœâš¿â™”æ—¥â, ç»´æš¿i¼ÆNginx â°°â° “â£ç + i¼Æsystemd æœâš¿ç°ç + ### ä, šæµ • æ¬šæ” “ç” Yç%©âÆ»â - |çš ‘æšæœ%é™â...¬â°, i½œç” Yç%©ä¿jæ°ç” â° “â • ¥ç”â, ^i½œ 2022.08 - 2024.07 - **â° ‘â¹³â°ç” Yä¿jâ • ¥â... • â¼äâ° “**i¼šR / Python / Linux / Docker / â° ‘âžYç” Y - **â, šâš¿è±âš”âÆ¬**i¼šâÆ»â¿è½¬â½ • ç»,ä, šâš¿æµç”è±âš”âÆ¬ i¼Ææ”jâ—âÆ¬â°â°â°ç” - **â¹³â°æ°â°°**i¼š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ### ä, šæµ • æ¬šæ” “ç” Yç%©âÆ»â¿æœ%é™â...¬â°, i½œç” Yç%©ä¿jæ°â°ž¹ç” Y i½œ 2021.10 - 2022.01 - â°ç” + èfžè½¬â½ • ç»,i¼šè ´”æžš æ°â, èš,,â + æžæ°â” ®âžžæ°ä, ææšâÆ¬â + æž - ä, »è¿æšæœ¬i¼šâ°ç” + èfžè½¬â½ • ç»,æ°Ræ°Pythonæ°Linux --- ## æ • ™è,²èfÆæ™- **è¥¿âÆ¬â + œæž—çš ‘æšæ°šâ¿i¼°æ°ç%©çš ‘â - |ä, žæšæœ¬ä, “ä, š i½œ æœ¬çš ‘i½œ 2018.09 - 2022.07 i½œ GPAi¼š3.35 - æ¬ • ä, šè°æ¬ ±i¼šâšâY°ä°žé»è¾ “æ¬¬è’ ,âžâ½’ çj°â°šâ°é°jâ€” ä¼ç», â...³ææšçš ¶â°šæš— â€” ææšçš ¶æÆ±æ± æ±â€¬ - â°šâ°éj¹ç»i¼šæ¹ç¬è°Ææž¥çš â¹ç´«èš±èœè “æš— é “œæ±jæY “çš,,â½±â “â°i¼^â°,ä, žä½œè€...i¼% --- ## ä, â°ä¼~âš¿ - **1 â “è±â¿K8s â°žç« - æ´**i¼šâ»žé¶â°v0.1.0 tagged éj¹ç»i¼Æç«¬ç«â°Ææ^â°Æ Helm chart + Python lib - **çž°æ°demo èf½âš, **i¼šâ°è±â°, |ç” µè,, ‘æœ” kind cluster ä, šè • ‘éšâ°ææ • ´ LLM æž”ç + æµç”i¼^éf”ç½²â + è - ç” ±â + chat completions â + metricsi¼% - **â¿â¹ èf½âš,â¼°**i¼šè±â¿ç” Yç%©ä¿jæ°æ°ç«—æ³ • æ°K8s ç - %â°âšéç + âYçY¥è¬ + i¼Æç«¬ç«â°Ææ^éj¹ç» - **3 â¹ Python â • ¥ç”âÆ¬ - ç»»é°Æ**i¼šä»žçš ‘ç” è,,šæœ¬â°â° ‘â¹³â°â • ¥â... • çš,,â°ææ • è - â¾,,